

DILSHA JAYASEKARA

Software Engineering Undergraduate

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🌐 [LinkedIn](#) | 💻 [GitHub](#) | 📁 [Portfolio](#) | 📄 [Medium](#)

SUMMARY

Third-year Software Engineering undergraduate at the University of Kelaniya with full-stack expertise in Java Spring Boot, the MERN stack, and microservices architecture. Proven track record of architecting secure e-commerce platforms and AI-powered distributed systems using Generative AI, RAG, and LangChain. Eager to leverage strong technical foundations to deliver scalable, high-performance, and intelligent software solutions.

EDUCATION

B.Sc.(Hons.) Software Engineering

University of Kelaniya

- CGPA: 3.61
- Specializing in Net-Centric Applications, Data Science and Engineering Applications, Health Informatic Applications domains.

G.C.E. Advanced Level

Telijjawila Central College

- Physical Science Stream
- z - score : 1.5045

SKILLS

Languages : Java, Python, JavaScript, TypeScript, PHP, C, HTML, CSS, SQL
Frontend : React.js, Next.js, Redux Toolkit, Tailwind CSS
Backend : Node.js, Express.js, Spring Boot, FastAPI, REST APIs
Databases : PostgreSQL, MongoDB, MySQL
AI & Data : Generative AI, LangChain, RAG Architecture, HuggingFace, Gemini API, Prompt Engineering
Tools & Platforms : Git, GitHub, Docker, Postman, Vercel, Power BI

PROJECTS

ELIX-Clothing-Store

MAR 2026 - MAY 2026

[LIVE SITE](#)

[GITHUB REPO](#)

Individual

- Developed a full-stack e-commerce platform using the **MERN** stack (**MongoDB**, **Express.js**, **React**, **Node.js**) for dynamic product catalogs and order processing.
- Secured backend API endpoints and session management with **JWT (JSON Web Tokens)** for user authentication and authorization.
- Integrated **Cloudinary API** for media management and optimized product image delivery.
- Managed global state for cart data and user authentication using **Redux Toolkit** for a responsive shopping experience.

Salona-Salon Management System | Ongoing

[GITHUB REPO](#)

Individual

- Architecting an enterprise **Microservices** platform (**Java Spring Boot**, **React.js**, **Python FastAPI**) routed via **API Gateway** with isolated **PostgreSQL** databases.
- Engineering an AI Receptionist for automated multi-step bookings using **LangChain**, **Google Gemini API**, and **Function Calling / Custom Tools**.
- Building a hallucination-free **RAG (Retrieval-Augmented Generation)** pipeline using **ChromaDB vector database** and **HuggingFace embeddings**.
- Integrating secure **RESTful APIs** and authentication across decoupled microservices, enforcing strict **DTO & Pydantic** validations for container-ready deployments.

Individual

- Built a scalable gaming platform backend using **microservices** with **Node.js** and **Express.js** for user management and authentication..
- Integrated backend **RESTful APIs** with a responsive, dynamic **React.js** frontend to ensure an interactive user experience.
- Designed robust data handling workflows using **MongoDB** to support high availability and secure user data management.
- Configured **inter-service communication** and **centralized error handling** to ensure fault tolerance across the distributed backend architecture.

Book Fair Reservation System

FEB 2026 – MAR 2026 [GITHUB REPO](#)

Academic – Group

- Engineered a full-stack stall reservation system for the Colombo International Book Fair using **Spring Boot** and **React (TypeScript)** to streamline vendor bookings.
- Implemented secure **JWT-based authentication** to manage user sessions and protect endpoints.
- Developed an automated **EmailService** workflow dispatching real-time booking confirmations and **QR code tickets** to vendors.
- Built an intuitive frontend vendor interface optimizing **state management** and **API integration** for a seamless booking workflow.

CERTIFICATES

- [Introduction to Cloud 101 – AWS Educate](#)
- [Getting Started to Storage – AWS Educate](#)

ACHIEVEMENTS

- Participated, Junior Hack hackathon (Software Engineering Students’ Association UOK)

VOLUNTEERING

- | | |
|---|-------------------------|
| • Software Engineering Students’ Association – Head Of Media Management | 📅 May 2026–Present |
| • Software Engineering Students’ Association – Committee member | 📅 April 2025 – May 2026 |
| • IEEEExtreme 19.0/KelaniXtreme 1.0 – Program Management Team | 📅 Sep 2025 – Oct 2025 |

LANGUAGES

- English (Professional Proficiency)
- Sinhala (Native)

REFERENCE

- **Prof. (Mrs) I.U. Hewapathirana**
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